

Memory Augmented Hierarchical Attention Network for Next Point-of-Interest Recommendation

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Abstract—Next point-of-interest (POI) recommendation has been an important task for location-based intelligent services. However, the application of such promising technique is still limited due to the following three challenges: 1) the difficulty of capturing complicated spatiotemporal patterns of user movements; 2) the hardness of modeling fine-grained long-term preferences of users; and 3) the effective learning of interaction between long- and short-term preferences. Motivated by this, we propose a memory augmented hierarchical attention network (MAHAN), which considers both short-term check-in sequences and long-term memories. To capture the complicated interest tendencies of users within a short-term period, we design a spatiotemporal self-attention network (ST-SAN). For long-term preferences modeling, we employ a memory network to maintain fine-grained preferences of users and dynamically operate them based on users' constantly updated check-ins. Moreover, we first employ a coattention network/mechanism to integrate the proposed ST-SAN and memory network, which can fully learn the dynamic interaction between long- and short-term preferences. Our extensive experiments on two publicly available data sets demonstrate the effectiveness of MAHAN.

Index Terms—Coattention, dynamic preference, memory network, point-of-interest (POI) recommendation, self-attention.

I. INTRODUCTION

LOCATION-BASED social networks (LBSNs) have become prevalent in our daily lives, e.g., Foursquare, Yelp, and Gowalla. In LBSNs, users would like to share their real-time geospatial locations and experiences on point of interests (POI) via check-in function. As a result, a large amount of accumulated check-in data are employed to mine personal preferences of users. Recently, next POI

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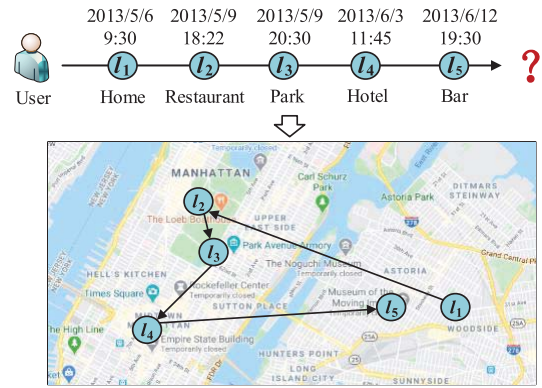


Fig. 1. Example of next POI recommendation.

recommendation has received extensive attention from both academia and industry [1]. As an extension of general POI recommendation [2]–[6], it aims to predict the POI a user tends to visit next based on his/her check-in sequence.

Various techniques have been proposed to model such sequential patterns. Early studies mainly employ Markov chain (MC) and matrix factorization (MF) to capture POI-to-POI transitions [7], [8]. These models rely on a strong assumption that a user's next behavior is only determined by the last check-in [9]. However, the next POI visited by a user is also highly correlated with other recently visited POIs [10]. Thus, some works leverage the word2vec framework [11] to learn embedding representations for POIs, which can reflect the contextual relationships of several continuously visited POIs [12], [13]. In addition, recurrent neural networks (RNNs) have demonstrated great power in sequence modeling tasks [14], [15], and some recent studies extend the original RNNs with various contextual information to model higher order sequential patterns of user movements [16]–[20].

Although current methods have achieved encouraging results, there are three remaining unresolved challenges. First, it is difficult to effectively model complicated spatiotemporal transition patterns within the check-in sequences. The main reason is that existing methods fail to fully explore the potential transition dependencies between any pair of visited POIs. As the example shown in Fig. 1, they only consider the transition relationship between the adjacent check-ins, e.g., l_1 to l_2 and l_2 to l_3 . However, the main factor affecting the user visits l_5 may due to the geographical proximity between l_5 and l_1 , rather than the influence of l_4 . Second,

it is hard to capture fine-grained user preferences in the long term. Users often show complicated interests in multiple aspects of POIs, e.g., categories, reputation, and location. However, existing methods often learn a static embedding vector [7], [21] or compress all of the user's check-in records into a fixed hidden vector [16], [18] to represent long-term preferences. Third, current models seldom consider interplay between long- and short-term preferences. It is noteworthy that both types of these two preferences are bidirectionally correlated, while existing studies often regard them as two independent modeling processes.

In this article, we propose a novel model called memory augmented hierarchical attention network (MAHAN) for next POI recommendation. It is able to capture users' dynamic preferences by combining short-term check-in sequences and long-term memories. To tackle the first challenge, we design a spatiotemporal self-attention network (ST-SAN) to capture short-term sequential patterns and interest tendencies of users. ST-SAN can effectively extend the vanilla self-attention network (SAN) [22] with temporal interval and geographical distance information of check-in sequences. Thus, complicated transition dependencies (i.e., semantic, temporal, and spatial relations) between any individual POI pairs can be thoroughly explored. To address the second challenge, we maintain users' multiple types of latent interests with key-value memory network (KV-MN) [23]. Comparing with the existing methods that merely adopt a compressed vector, our approach strengthens the representation capacity in modeling fine-grained long-term preferences. In addition, a novel read/write operation is designed to dynamically operate the memory preferences of users based on their updated check-in records. Finally, we design a coattention network, which combines ST-SAN and KV-MN to model the dynamic interaction between long- and short-term preferences. The proposed model can obtain high-level user representations with co-dependent and mutually enhanced long- and short-term preferences. Specifically, the contributions of this article are as follows.

- 1) We propose a hybrid neural model, MAHAN, to capture both dynamic long- and short-term preferences. Moreover, a coattention network is designed to model the interplay between these two types of preferences.
- 2) We design an ST-SAN for short-term preferences learning. ST-SAN can fully leverage spatiotemporal context information to explore complicated and subtle transition dependencies within the check-in sequences.
- 3) We utilize memory networks to store fine-grained long-term preferences of users at the feature level. In addition, memory preferences can be dynamically visited and updated based on users' up-to-date check-ins.
- 4) Extensive experiments are conducted on two real-world LBSN data sets, and results demonstrate that our model consistently outperforms the state-of-the-art methods in different evaluation metrics.

The remainder of this article is organized as follows. In Section II, we first briefly review the related work. Then, we present the problem formulation in Section III. Section IV introduces the detail of the proposed model - MAHAN.

The experimental results are shown in Section V. Finally, we conclude this work in Section VI.

II. RELATED WORK

In this section, we briefly review the most relevant research on general POI recommendation, next POI recommendation, and attention mechanism for recommendation system.

A. General POI Recommendation

POI recommendation has become an important topic in many location-based applications. Early studies mainly focused on capturing users' general preferences from user-POI interaction records without considering the order of check-ins. Among these studies, collaborative filtering (CF), such as MF, is one of the most effective methods [24]. However, directly applying CF to POI recommendation suffers an issue of data sparsity. Therefore, many studies integrated CF with multiple context information to alleviate this problem [2]–[5], [13], [25]. For example, Cheng *et al.* [3] fused social and geographical influence into a basic MF framework. Li *et al.* [4] proposed a ranking-based factorization method to incorporate different types of context information. Recently, deep learning was applied to model higher level user-POI relationships [1], [6], [26]. For example, Ma *et al.* [26] proposed an autoencoder-based model that consists of a self-attentive encoder and a neighbor-aware decoder to learn nonlinear user-POI interactions. Zhou *et al.* [6] proposed a topic-enhanced memory network to capture both global and fine-grained user preferences with different spatiotemporal contexts.

B. Next POI Recommendation

Different to general POI recommendation, next POI recommendation pays more attention to model sequential patterns of historical check-ins and dynamic evolution of user preferences [10], [17]. Early studies assumed that the transition relations between POIs follow the first-order MC. For instance, Cheng *et al.* [7] proposed an MF method (FPMC-LR) to combine the personalized MC in the check-in sequence and localized region constraint. Feng *et al.* [8] proposed a personalized ranking metric embedding method (PRME-G) to jointly model sequential information and individual preferences. Inspired by the success of word2vec [11] in natural language processing (NLP), Feng *et al.* [21] proposed an embedding model (POI2Vec) to incorporate the geographical influence of POIs into the framework of word2vec. This model can also jointly learn user preference and POI sequential transition. However, these methods can only capture point-level transition relationships among POIs.

Recently, many works focus on modeling complex sequential patterns by integrating various contextual information into the original RNNs. For example, Liu *et al.* [16] proposed a spatial-temporal recurrent neural network (ST-RNN) by extending the classic RNN with time- and distance-special transition metrics. Li *et al.* [18] and Huang *et al.* [27] incorporated multiple contextual information associated with check-ins into LSTM to better understand sequential transitions of

user movements better. Zhao *et al.* [28] designed a spatiotemporal gated network (STGN) to capture the spatiotemporal relationships between the adjacent check-ins. Sun *et al.* [10] introduced a hybrid model (LSTPM) by designing a geodilated RNN for short-term preference learning and a context-aware nonlocal network for long-term preference modeling. Although RNN-based methods have demonstrated effectiveness and scalability in sequential prediction problems, one major issue is that they often encode the overall check-in sequence into a hidden state without distinguishing the significance of each check-in for prediction [29]. In addition, they only model the global sequence dependence and thus fail to explicitly learn subtle transition relationships between any individual POI pairs [9].

C. Attention Mechanism for Recommendation System

The neural attention mechanism is introduced to dynamically capture the important parts of an input. It has been widely applied in diverse applications, such as NLP [14], computer vision [15], and recommendation system [27], [30]. The original vanilla attention mechanism is often integrated into RNN-based methods, which can memorize long-range dependencies or focus on important information of sequences [18], [19]. For example, Feng *et al.* [19] designed an attentional recurrent model (DeepMove) that used the short-term status to selectively focus on relevant historical records.

Recently, self-attention [22] has been successfully applied in machine translation. As a special case of the vanilla attention mechanism, it shows higher effectiveness and efficiency in sequence modeling than RNN and CNN. Thus, many studies have applied it in general sequential recommendations [9], [30], [31]. For example, Kang *et al.* [30] designed a sequential model (SASRec) based on SANs, which can capture long-term sequential dependence and identify relevant historical behaviors to predict the next items. Zhang *et al.* [32] proposed a hybrid model (AttRec) to utilize self-attention to infer item-item relationships and model short-term user intents. Its integration with long-term preferences is achieved through a metric learning framework. Although those existing self-attention-based methods have achieved good performance in sequential recommendation, they cannot be directly used in next POI recommendation. The major concern is that they neglect complicated spatiotemporal transition patterns of the check-in sequences. In this article, we improve the SAN by incorporating spatiotemporal check-ins information, in order to model dynamic sequential-transition dependencies of users.

III. NOTATIONS AND PROBLEM FORMULATION

Table I presents the notations used in this article. Let $U = \{u_1, u_2, \dots, u_{|U|}\}$ denote a set of users and $L = \{l_1, l_2, \dots, l_{|L|}\}$ represent a set of POIs, where $|U|$ and $|L|$ are the total numbers of users and POIs, respectively. For the ease of illustration, we give the following definitions.

Definition 1 (POI): A POI is a spatial item (e.g., a restaurant and a museum) with unique geographical coordinates (latitude and longitude). In this article, it is described by a triple: $\langle l_{id}, \text{lat}_{l_{id}}, \text{lon}_{l_{id}} \rangle$.

TABLE I
NOTATIONS USED IN THIS ARTICLE

Notation	Description
U, L	sets of users and POIs
c_t^u	check-in record of user u
H^u	historical check-in sequence of user u
S_{t-1}^u	short-term check-in sequence of user u
e_l	embedding of POI l
$\mathbf{p}^u$	preference representation of user u
$\mathbf{E}^u$	short-term sequence embedding matrix
$\mathbf{F}^u$	output representation of short-term model
$\mathbf{K}$	key matrix of memory network
$\mathbf{V}^u$	value matrix of memory network
$\mathbf{S}$	similarity matrix
α	attention weight vector of model
$\{\mathbf{W}\}, \{\mathbf{b}\}$	weight metrics and bias vectors of model
k	length of short-term sequence
m	number of memory slots
τ	maximum time interval
h	number of heads in self-attention network

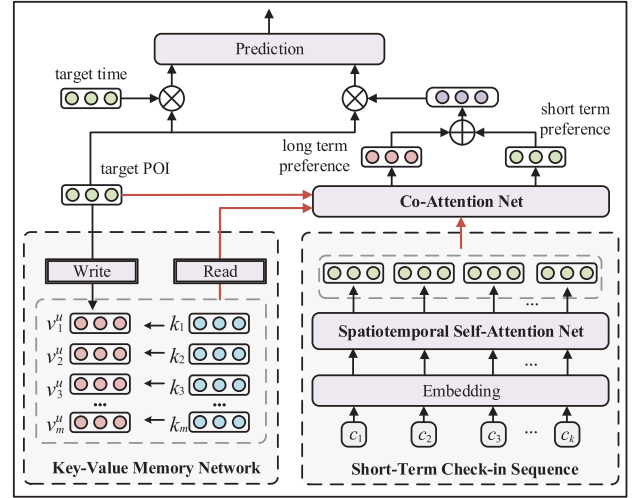


Fig. 2. Overall architecture of MAHAN.

Definition 2 (Check-In): A user's check-in record is represented by a triple $c_t^u = \langle u, l_t^u, t \rangle$, which means that a user $u \in U$ visited a POI $l_t^u \in L$ at time t .

Definition 3 (Check-In Sequence): A check-in sequence of a user u is his/her historical check-in records in chronological order, denoted by $H^u = \{c_{t_1}^u, c_{t_2}^u, \dots, c_{t_{|H^u|}}^u\}$, where $|H^u|$ represents the total number of check-ins. Specifically, given a specific time t , we take the most recent k check-ins before t as the short-term check-in sequence, denoted by $S_{t-1}^u = \{c_{t-k}^u, c_{t-k+1}^u, \dots, c_{t-1}^u\}$, which can reflect short-term preferences of users.

Problem Statement: Given the check-in sequence H^u of user u and a query time t_q , we aim to predict the probability of an unvisited POI $l_{t_q} \in L/H^u$ that user u will visit at time t_q . Finally, we will recommend a ranking list of top-N POIs for the target user according to the predicted probability scores.

IV. PROPOSED MODEL

In this section, the proposed MAHAN model is introduced. The overall architecture of MAHAN is presented in Fig. 2. It consists of three core components: ST-SAN,

KV-MN, and coattention network. The ST-SAN models complicated spatiotemporal transition patterns within the check-in sequence and captures the short-term preferences of users. The KV-MN leverages external memories to maintain fine-grained and dynamic long-term preferences of users in the feature level. ST-SAN and KV-MN are closely linked with each other through the coattention network, which models the dynamic interaction between long- and short-term preferences. In the following, we introduce the design of MAHAN in detail and illustrate how to jointly train the model.

A. Spatiotemporal Self-Attention Network

Users' check-in records often involve complicated spatial and temporal transition patterns, which are critical for better understanding of users' sequential preferences. Thus, we propose incorporating spatiotemporal information of check-in sequences into the vanilla SAN for modeling short-term spatiotemporal transition dependencies.

Given a specific time t , we obtain the short-term sequence S_{t-1}^u of user u and take it as the input of the short-term module. With an embedding layer, every POI $l_i^u \in S_{t-1}^u$ is converted into a d -dimensional vector $\mathbf{e}_i^u \in \mathbb{R}^d$ and we can retrieve the embedding representation of short-term sequence $\mathbf{E}^u = \{\mathbf{e}_1^u, \mathbf{e}_2^u, \dots, \mathbf{e}_k^u\} \in \mathbb{R}^{d \times k}$. To model the sequence, we conduct a standard multihead self-attention on $\mathbf{E}^u$ similar with previous works [30], [31] and obtain a new sequence representation $\mathbf{A}^u \in \mathbb{R}^{d \times k}$

$$\begin{aligned} \mathbf{A}^u &= \text{MultiHeadAttention}(\mathbf{E}^u) \\ &= \mathbf{W}_o[\text{head}^1 \oplus \text{head}^2 \oplus \dots \oplus \text{head}^h] \end{aligned} \quad (1)$$

where $\mathbf{W}_o \in \mathbb{R}^{d \times h d_o}$ is a linear projection matrix, h is the number of heads, $d_o = d/h$, and $\oplus$ is the concatenation operation. Specifically, $\text{head}^\ell = \{\mathbf{a}_1^\ell, \mathbf{a}_2^\ell, \dots, \mathbf{a}_k^\ell\} \in \mathbb{R}^{d_o \times k}$ is the output of ℓ th head, and each output element $\mathbf{a}_i^\ell$ is the weighted combination of POI embeddings of itself and other contextual POIs in the short-term sequence

$$\mathbf{a}_i^\ell = \sum_{j=1}^k \alpha_{i,j} (\mathbf{W}_v \mathbf{e}_j^u). \quad (2)$$

The attention weight $\alpha_{i,j}$ that measures the semantic similarity between POIs is calculated as

$$\alpha_{i,j} = \text{softmax} \left(\frac{(\mathbf{W}_Q \mathbf{e}_i^u)^T \mathbf{W}_K \mathbf{e}_j^u}{\sqrt{d}} \right) \quad (3)$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d}$ are linear transformation metrics. However, the vanilla self-attention treat the check-in sequence as a simple word sequence of NLP without considering any spatial and temporal relationships among the check-ins.

1) *Modeling Temporal Transition Patterns*: In order to model temporal transition patterns of the check-in sequence, the time interval between any pair of check-ins is considered.

We obtain a temporal relationship matrix of the check-in sequence $\mathbf{R}_t \in \mathbb{R}^{k \times k}$ as

$$\mathbf{R}_t = \begin{bmatrix} t_{1,1} & t_{1,2} & \dots & t_{1,k} \\ t_{2,1} & t_{2,2} & \dots & t_{2,k} \\ \vdots & \vdots & \ddots & \vdots \\ t_{k,1} & t_{k,2} & \dots & t_{k,k} \end{bmatrix} \quad (4)$$

where $t_{i,j} = t_j - t_i$ denotes the time interval between the i th and j th check-ins. Moreover, $t_{i,j} = -t_{j,i}$. It means that bidirectional and relative temporal relationships among check-ins can be explored. Then, we discretize the time interval by day and convert every time interval $t_{i,j}$ into an embedding vector $\mathbf{r}_{i,j} \in \mathbb{R}^d$ by

$$\mathbf{r}_{i,j} = \mathbf{w}_{\text{clip}(t_{i,j}, \tau)} \quad (5)$$

$$\text{clip}(x, \tau) = \max(-\tau, \min(\tau, x)) \quad (6)$$

where τ is the width of time window. As accurate relative time information is not helpful beyond a certain time interval, we clip the maximum relative time interval to a maximum absolute value of τ . We additionally learn $2\tau + 1$ unique relative temporal representations $\mathbf{w} = (\mathbf{w}_{-\tau}, \dots, \mathbf{w}_\tau)$, where $\mathbf{w}_i \in \mathbb{R}^d$. Considering relative temporal information, the standard self-attention in (2) is modified as

$$\mathbf{a}_i^\ell = \sum_{j=1}^k \alpha_{i,j} (\mathbf{W}_{V,E} \mathbf{e}_j^u + \mathbf{W}_{V,R} \mathbf{r}_{i,j}) \quad (7)$$

$$\alpha_{i,j} = \text{softmax} \left(\frac{(\mathbf{W}_Q \mathbf{e}_i^u)^T (\mathbf{W}_{K,E} \mathbf{e}_j^u + \mathbf{W}_{K,R} \mathbf{r}_{i,j})}{\sqrt{d}} \right) \quad (8)$$

where $\mathbf{W}_{K,E}, \mathbf{W}_{K,R} \in \mathbb{R}^{d \times d}$, and $\mathbf{W}_{V,E}, \mathbf{W}_{V,R} \in \mathbb{R}^{d \times d}$ are linear transformation metrics. In this way, temporal dynamic dependencies of users' check-in histories are modeled.

2) *Modeling Spatial Transition Patterns*: Users' check-ins often show obvious geographical clustering phenomenon that is geographically nearby POIs are more likely to be visited in succession [26], [33]. To model spatial transition patterns, we obtain a geographical relationship matrix $\mathbf{R}_d \in \mathbb{R}^{k \times k}$ similar with temporal patterns modeling, where each element $d_{i,j} \in \mathbf{R}_d$ denotes the geographical distance between the POI l_i and the POI l_j , i.e., $d_{i,j} = ((\text{lon}_i - \text{lon}_j)^2 + (\text{lat}_i - \text{lat}_j)^2)^{1/2}$. Then, we adopt the Gaussian radial basis function (RBF) kernel to further incorporate the influence of geographical distance [26], and the RBF kernel is defined as: $g_{i,j} = \exp(-\gamma \|d_{i,j}\|^2)$, where γ is a scaling parameter that controls the geographical correlation degree of two POIs. The calculation of attention weight $\alpha_{i,j}$ in (8) is updated to

$$\alpha_{i,j} = \text{softmax} \left(\frac{(\mathbf{W}_Q \mathbf{e}_i^u)^T (\mathbf{W}_{K,E} \mathbf{e}_j^u + \mathbf{W}_{K,R} \mathbf{r}_{i,j})}{\sqrt{d}} + \eta \cdot g_{i,j} \right) \quad (9)$$

where η is a weighing parameter. So far, we have designed a spatiotemporal-aware self-attention, and complicated transition dependencies (i.e., semantic, temporal, and spatial relations) among POIs can be modeled.

To enhance the model with nonlinearity, we employ a pointwise feedforward network on $\mathbf{A}^u$ as

$$\mathbf{A}^u = \text{LayerNorm}(\mathbf{A}^u + \mathbf{E}^u) \quad (10)$$

$$\mathbf{F}^u = \mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{A}^u + \mathbf{b}_1) + \mathbf{b}_2 \quad (11)$$

$$\mathbf{F}^u = \text{LayerNorm}(\mathbf{F}^u + \mathbf{A}^u) \quad (12)$$

where $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{d \times d}$ are the weight matrices and $\mathbf{b}_1, \mathbf{b}_2 \in \mathbb{R}^d$ are the bias vectors. Moreover, residual connection and layer normalization are also employed to learn meaningful features hierarchically and stabilize model training. The output $\mathbf{F}^u = \{\mathbf{h}_1^u, \mathbf{h}_2^u, \dots, \mathbf{h}_k^u\} \in \mathbb{R}^{d \times k}$ is the final representation of short-term sequence. How to calculate short-term preferences from $\mathbf{F}^u$ will be introduced in Section III-C.

B. Key-Value Memory Network

Existing methods usually encode long-term preferences of users into a static latent vector, which neglects the dynamic and diverse properties of long-term preferences. In this work, we employ memory networks [6], [23] to preserve users' multiple types of latent interests and manipulate them in a dynamic way.

KV-MN [23] utilizes a large number of external memory slots to explicitly store rich information. It has been proved to be capable of effectively modeling long-term sequential data [29], [34]. In KV-MN, a memory slot is composed of a key vector and an associated value vector. Suppose that there are a set of latent features that shared by all POIs and users' preferences are affected by these features. We define $\mathbf{M}^u = (\mathbf{K}, \mathbf{V}^u)$ as the long-term memory of user u .

Specifically, $\mathbf{K} = \{\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_m\} \in \mathbb{R}^{d \times m}$ is a key matrix composed of the embeddings of latent features, m is the number of latent features, and $\mathbf{k}_i \in \mathbb{R}^d$ represents the embedding of feature i . $\mathbf{V}^u = \{\mathbf{v}_1^u, \mathbf{v}_2^u, \dots, \mathbf{v}_m^u\} \in \mathbb{R}^{d \times m}$ is a value matrix to store users' preference characteristics for corresponding features, and $\mathbf{v}_i^u$ is the representation of u 's preference on feature i . The key matrix $\mathbf{K}$ is shared by all users, whereas the value matrix $\mathbf{V}^u$ is set to be private for each user since users often own their individual preferences over the shared features. To model personal and time-evolving preferences of users, we dynamically revise the value matrix for each user based on their constantly updated check-ins. Therefore, two types of operations are designed to operate the memory network.

1) *Read Operation*: Given a user-POI pair $\langle u, l \rangle$, the user's memory embeddings of different features are not equally important when predicting u 's preference on the target POI l . We use the embedding of target POI $\mathbf{e}_l$ as the query to attentively visit the user's memory network. The attention weight $\alpha_k \in \mathbb{R}^m$ is calculated by

$$\begin{aligned} \alpha_k &= \mathbf{V}_k^T \text{ReLU}(\mathbf{W}_k \mathbf{K} + \mathbf{W}_l \mathbf{e}_l + \mathbf{b}_k) \\ \alpha_k &= \text{softmax}(\alpha_k) \end{aligned} \quad (13)$$

where $\mathbf{W}_k, \mathbf{W}_l \in \mathbb{R}^{d \times d}$, $\mathbf{V}_k \in \mathbb{R}^d$ are the weight matrices and $\mathbf{b}_k \in \mathbb{R}^d$ is a bias vector. Finally, the user's long-term preference $\mathbf{p}_m^u$ can be formulated as the weighted combination

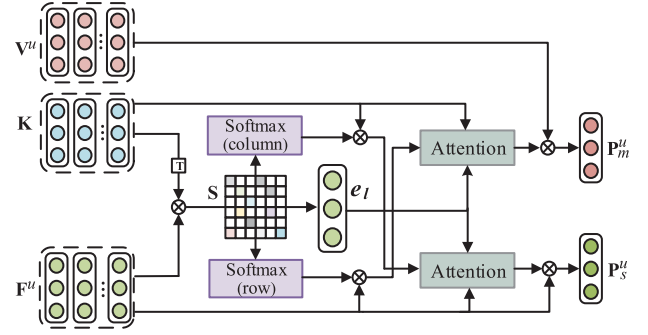


Fig. 3. Structure of coattention network.

of value vectors in $\mathbf{V}^u$

$$\mathbf{p}_m^u = \sum_{i=1}^m \alpha_{k,i} \mathbf{v}_i^u. \quad (14)$$

2) *Write Operation*: A user's preference tends to be dynamically influenced and reflected by the POIs that he/she visited. Through the write operation, we continually update the value matrix $\mathbf{V}^u$ according to the user's check-in records. Motivated by the design of different gates in LSTM [35], we first drive a gate vector $\mathbf{g}_i \in \mathbb{R}^d$ for each value vector $\mathbf{v}_i^u$ to control what information in $\mathbf{v}_i^u$ should be erased and what information should be updated from $\mathbf{e}_l$ by

$$\mathbf{g}_i = \sigma(\mathbf{W}_v \mathbf{v}_i^u + \mathbf{W}_l \mathbf{e}_l + \mathbf{b}_g) \quad (15)$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the sigmoid function and $\mathbf{W}_v \in \mathbb{R}^{d \times d}$ is the weight matrix. With the gate vector, we update each value vector $\mathbf{v}_i^u$ by

$$\mathbf{v}_i^u \leftarrow \mathbf{g}_i \odot \mathbf{v}_i^u + (\mathbf{1} - \mathbf{g}_i) \odot \mathbf{e}_l \quad (16)$$

where $\odot$ is the elementwise multiplication. This update operation allows our model dynamically maintain fine-grained long-term preferences of users in the feature level.

C. Coattention Network

The potential correlation and mutual effect between long- and short-term preferences are often overlooked. Inspired by the studies of vision-language tasks [15], [36], we design a symmetrical coattention network that consists of memory-based sequence attention and sequence-based memory attention. In this way, our model can capture codependent and mutually enhanced long- and short-term preference representations. The coattention network is shown in Fig. 3.

Taking the long-term memory $\mathbf{M}^u$ and short-term sequence representation $\mathbf{F}^u$ as the input of the coattention network, we first construct a similarity matrix $\mathbf{S} \in \mathbb{R}^{m \times k}$ through

$$\mathbf{S} = \mathbf{K}^T \mathbf{W}_s \mathbf{F}^u \quad (17)$$

where $\mathbf{W}_s \in \mathbb{R}^{d \times d}$ is a weight matrix. Each entry of $\mathbf{S}$ measures the similarity between each feature in the long-term memory and each POI in the short-term sequence. With the similarity matrix $\mathbf{S}$, we conduct the coattention separately on the short-term sequence and the long-term memory.

1) *Memory-Based Sequence Attention*: Memory-based sequence attention aims to find the most important POIs in the sequence from the perspective of long-term preferences. We normalize $\mathbf{S}$ in columnwise to derive an attention map on memory features conditioned by each POI in the sequence: $\mathbf{S}_k = \text{softmax}_{\text{col}}(\mathbf{S})$. Then, we apply the attention map $\mathbf{S}_k$ to key matrix $\mathbf{K}$ and obtain the attended memory features: $\mathbf{A}_k = \mathbf{K}\mathbf{S}_k$. We measure the importance of each POI in $\mathbf{S}_{t-1}^u$ by considering the influence of both long-term preference and target POI, and the attention weight is calculated by

$$\begin{aligned}\alpha_s &= \mathbf{V}_s^T \text{ReLU}(\mathbf{W}_s \mathbf{F}^u + \mathbf{W}_{ak} \mathbf{A}_k + \mathbf{W}_l \mathbf{e}_l + \mathbf{b}_s) \\ \alpha_s &= \text{softmax}(\alpha_s)\end{aligned}\quad (18)$$

where $\mathbf{W}_s, \mathbf{W}_{ak} \in \mathbb{R}^{d \times d}$, $\mathbf{V}_s \in \mathbb{R}^d$ are the weight matrices and $\mathbf{b}_s \in \mathbb{R}^d$ is a bias vector. The short-term preference $\mathbf{p}_s^u \in \mathbb{R}^d$ is formulated as the weighted sum of fused POI embeddings in the short-term sequence

$$\mathbf{p}_s^u = \sum_{i=1}^k \alpha_{s,i} \mathbf{h}_i^u. \quad (19)$$

2) *Sequence-Based Memory Attention*: Conversely, sequence-based memory attention aims to identify which features in the long-term memory are most relevant to each POI in the short-term sequence. Similar to operation of memory-based sequence attention, we normalize $\mathbf{S}$ in rowwise to obtain the attention map on POIs for each memory feature: $\mathbf{S}_f = \text{softmax}_{\text{row}}(\mathbf{S})$ and apply it to $\mathbf{F}^u$ to obtain the attended sequence features: $\mathbf{A}_s = \mathbf{F}^u \mathbf{S}_f^T$. By taking the influence of short-term preference into consideration, the calculation of attention weight α_k in (13) is replaced with

$$\begin{aligned}\alpha_k &= \mathbf{V}_k^T \text{ReLU}(\mathbf{W}_k \mathbf{K} + \mathbf{W}_{as} \mathbf{A}_s + \mathbf{W}_l \mathbf{e}_l + \mathbf{b}_k) \\ \alpha_k &= \text{softmax}(\alpha_k)\end{aligned}\quad (20)$$

where $\mathbf{W}_{as} \in \mathbb{R}^{d \times d}$ is the weight matrix. We calculate the user's long-term preference with (14). In this way, our model can effectively manipulate the memory network by taking short-term preferences into consideration. Moreover, every module of MAHAN is linked with each other, resulting in an end-to-end POI recommendation framework.

D. Prediction and Training

After obtaining the long-term preference $\mathbf{p}_m^u$ and short-term preference $\mathbf{p}_s^u$ of user u , we combine these two d -dimensional preference vectors and get a more well-rounded user representation as follows:

$$\mathbf{p}^u = \mathbf{W}_p [\mathbf{p}_m^u \oplus \mathbf{p}_s^u] + \mathbf{b}_p \quad (21)$$

where $\oplus$ is the concatenation operator, $\mathbf{W}_p \in \mathbb{R}^{d \times 2d}$ is the weight matrix, $\mathbf{b}_p \in \mathbb{R}^d$ is the bias vector, and $\mathbf{p}^u \in \mathbb{R}^d$ is the user's final preference representation.

Given a specific prediction request $q = \langle u, t, l_i \rangle$, we consider the influence of user preference and POI temporal property simultaneously to calculate the overall prediction score through the following equation:

$$s_{u,t,i} = \mathbf{e}_i^T \mathbf{p}^u + \mathbf{e}_i^T \mathbf{e}_t \quad (22)$$

TABLE II
STATISTICS OF TWO DATA SETS

Datasets	Foursquare	Gowalla
#Users	23,686	20,364
#POIs	27,055	28,776
#Check-ins	1,121,315	1,700,250
Avg. check-ins per user	47.34	83.49
Avg. check-ins per POI	41.45	59.86
Density	0.17%	0.29%

where $\mathbf{e}_i$ and $\mathbf{e}_t$ are the embeddings of POI l_i and time t , respectively. In the prediction function, the first term models user's preference on the POI and the second term captures the influence of time.

In general, the task of next POI recommendation can be regarded as a binary classification problem. Thus, we adopt the binary cross entropy as the loss function and let the regularized log-likelihood function as the objective function

$$\begin{aligned}\mathcal{L}(\Theta) &= - \sum_{u \in U} \left(\sum_{i \in H^u} \log(\sigma(s_{u,t,i})) + \sum_{j \notin H^u} \log(1 - \sigma(s_{u,t,j})) \right) \\ &\quad + \frac{\lambda}{2} \|\Theta\|^2\end{aligned}\quad (23)$$

where λ is the regularization parameter. Since only positive samples exist in the data set, it is intractable to directly train the model. Following previous works [9], [30], for each target POI l_i of u , we randomly sample several (1 in our work for simplicity) POIs that the user u has not visited as negative instances. $\Theta = \{\mathbf{X}, \mathbf{M}, \mathbf{T}, \mathbf{w}, \mathbf{W}, \mathbf{b}\}$ is the model parameter set. All the model parameters are learned by minimizing the objective function $\mathcal{L}$ on the training data. We adopt the adaptive moment estimation (Adam) optimizer to automatically adjust the learning rate during the training procedure. In our model, we also employ the dropout for fully connected layers to avoid overfitting problem.

V. EXPERIMENTS

In this section, we first introduce the experimental setup and then compare the performance of our model with start-of-the-art baselines.

A. Experimental Setup

1) *Data Sets*: We conduct experiments on two publicly available LBSN data sets: Foursquare¹ and Gowalla.² The Foursquare data set is provided in [37], which collected check-ins of users in USA from February 2012 to September 2013. The Gowalla data set provided in [25] is a larger data set, and we take a subset that was generated worldwide from January 2011 to August 2011 for experiments. Each check-in record in the data sets contains a user-ID, a POI-ID, a timestamp, and the latitude and longitude of this POI. For both data sets, we discard users with fewer than ten check-ins and POIs with fewer than ten visitors. The statistics of data sets are shown in Table II. Given a user's check-in

¹<http://spatialkeyword.sce.ntu.edu.sg/eval-vldb17>

²http://www.yongliu.org/data_sets/index.html

TABLE III
OVERALL PERFORMANCE COMPARISON OF ALL METHODS IN TERMS OF ACC@N, NDCG@N, AND MRR

Dataset	Methods	Acc@5	Acc@10	Acc@20	NDCG@5	NDCG@10	NDCG@20	MRR
Foursquare	POP	0.0833	0.1028	0.1255	0.0669	0.0732	0.0789	0.0632
	FPMC-LR	0.1269	0.1625	0.2077	0.0989	0.1104	0.1217	0.0914
	PRME-G	0.1301	0.1685	0.2175	0.1003	0.1127	0.1250	0.0990
	POI2Vec	0.1161	0.1523	0.1994	0.0888	0.0984	0.1123	0.0882
	ST-RNN	0.1406	0.1781	0.2275	0.1120	0.1241	0.1365	0.1092
	RUM	0.1412	0.1747	0.2220	0.1108	0.1216	0.1335	0.1083
	SASRec	0.1458	0.1842	0.2347	0.1148	0.1268	0.1385	0.1148
	AttRec	0.1467	0.1884	0.2411	0.1143	0.1277	0.1410	0.1152
	MAHAN	0.1653	0.2082	0.2662	0.1254	0.1391	0.1533	0.1289
Gowalla	POP	0.0872	0.1242	0.1670	0.0705	0.0823	0.0956	0.0632
	FPMC-LR	0.1151	0.1551	0.2041	0.0852	0.0981	0.1104	0.0856
	PRME-G	0.1191	0.1633	0.2131	0.0856	0.0998	0.1124	0.0868
	POI2Vec	0.1082	0.1658	0.2214	0.0775	0.0961	0.1151	0.0846
	ST-RNN	0.1215	0.1692	0.2349	0.0865	0.1018	0.1183	0.0891
	RUM	0.1224	0.1705	0.2342	0.0885	0.1009	0.1199	0.0914
	SASRec	0.1336	0.1820	0.2468	0.0956	0.1074	0.1256	0.0918
	AttRec	0.1323	0.1858	0.2584	0.0932	0.1116	0.1289	0.0962
	MAHAN	0.1496	0.2055	0.2889	0.1068	0.1247	0.1437	0.1091

sequence, the most recent check-in is hold out for testing the model's performance, the second most check-in is used as the validation data, and all the remaining check-ins are left for training the model.

2) *Compared Methods*: We compare MAHAN with the following state-of-the-art recommendation models.

- 1) *POP*: This is a nonpersonalized method, and all POIs are recommended based on their popularity that is determined by the number of visitors.
- 2) *FPMC-LR* [7]: This method combines MF and personalized MC to model user preference and sequential information. It also considers the spatial movement constraints of users.
- 3) *PRME-G* [8]: This is a metric embedding model that jointly learns sequential transition and individual preference, and it integrates geographical influence through the weight of geographical distances.
- 4) *POI2Vec* [21]: This is an embedding model that fuses geographical relations of POIs into the word2vec framework and aggregates POI sequential transition to predict the next POIs.
- 5) *ST-RNN* [16]: This is an RNN-based next POI recommendation model, which integrates time interval and geographical distance information between two consecutive check-ins into the classic RNN.
- 6) *RUM* [29]: This method leverages memory networks integrated with CF to preserve and update users' preferences for sequential recommendation.
- 7) *SASRec* [30]: This is a self-attention-based sequential recommendation model, which uses an SAN to model user behavior sequence and adaptively identify relevant behaviors for predictions.
- 8) *AttRec* [32]: It employs a self-attention module to capture short-term interests and a metric learning component to model long-term preferences.

3) *Evaluation Metrics*: To evaluate the performance of different models, we adopt three widely used metrics: Acc@N, NDCG@N, and MRR. Acc@N is defined as the percentage

of test POIs that can appear in the top-N recommended POIs. We calculate Acc@N as proposed in [9]. NDCG@N measures the ranking quality by considering the position of the correctly recommended POI. MRR is a popular ranking metric in the information retrieval field, and we use it to measure the average ranking of ground-truth POIs [32].

4) *Implementation Details*: We implement our MAHAN model with Python 3.6 and Pytorch 1.3 running on GPU machines of Nvidia GeForce GTX 2080 Ti.³ Optimizer is the Adam optimizer. The initial learning rate is set to 0.001, the regularization coefficient is 5×10^{-5} , and the dropout rate is 0.5. The embedding size of all the models is set to 50 for fair comparison. For parameters in the short-term module, we set the scaling parameter $\gamma = 10$ and the weighing coefficient $\eta = 0.6$ for both data sets. The width of relative time window τ is set to 5 and 3 for Foursquare and Gowalla, respectively. The short-term sequence length k is set to 6 for Foursquare and 4 for Gowalla. We set the number of head $h = 2$ for both data sets. For the number of memory slot m , $m = 5$ and $m = 10$ give the best performances for Foursquare and Gowalla, respectively.

B. Overall Performance Comparison

The overall performance of different models on two data sets is shown in Table III. From the experimental results, we have the following observations.

First, the proposed MAHAN significantly and consistently outperforms all the compared methods on both data sets in terms of all evaluation metrics. Specifically, on the Foursquare data set, MAHAN achieves up to 10.4%, 8.7%, and 11.8% improvements compared with the best baseline (i.e., AttRec) in terms of Acc@20, NDCG@20, and MRR, respectively, whereas the improvements on the Gowalla data set are 11.8%, 11.5%, and 13.4% in the same metrics compared with the second best result. This demonstrates the feasibility and effectiveness of our model in capturing users' dynamic preferences. Compared with two self-attention based models

³Code is available at https://github.com/guolipa/MAHAN_TCSS

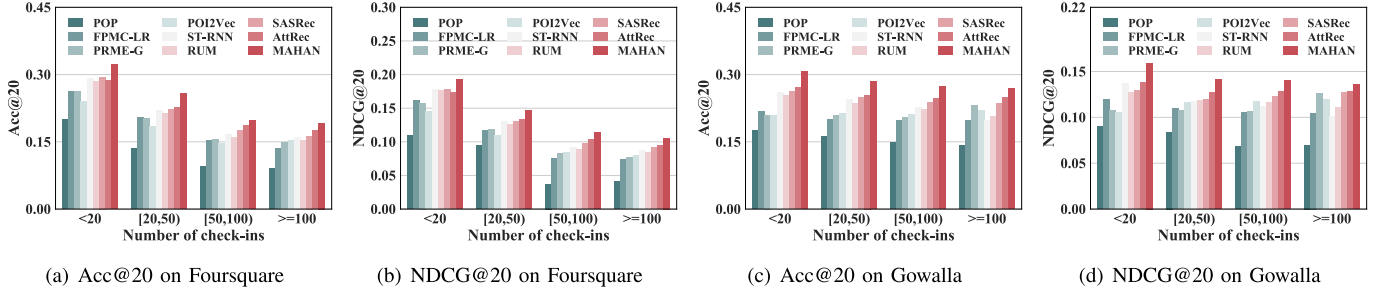


Fig. 4. Performance on different numbers of historical check-ins. (a) Acc@20 on Foursquare. (b) NDCG@20 on Foursquare. (c) Acc@20 on Gowalla. (d) NDCG@20 on Gowalla.

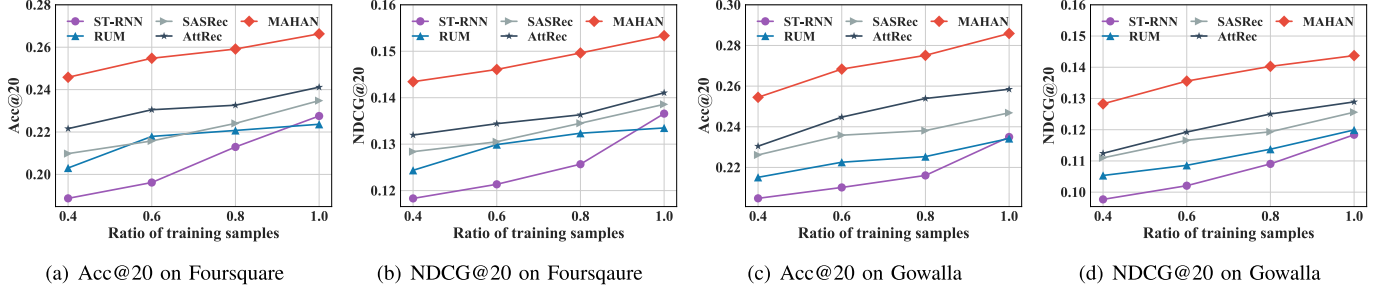


Fig. 5. Performance on different ratios of training samples. (a) Acc@20 on Foursquare. (b) NDCG@20 on Foursquare. (c) Acc@20 on Gowalla. (d) NDCG@20 on Gowalla.

(i.e., AttRec and SASRec), the superiority of MAHAN stems from three reasons. First, MAHAN improves the vanilla self-attention to model complicated spatiotemporal patterns of the sequences. Second, MAHAN can model fine-grained long-term preferences in a dynamic way, whereas the baselines only use a static embedding vector. The third reason is the coattention network that jointly models dynamic interaction between long- and short-term preferences, while other methods fail to do so.

Second, considering the baseline methods, deep learning-based methods achieve higher prediction accuracy than classical methods. The nonpersonalized model POP gives the worst performance, and this highlights the importance of providing personalized recommendations for users. ST-RNN performs better than the MC-based method (i.e., FPMC-LR) and embedding-based methods (i.e., PRME-G and POI2Vec). The main reason is that these methods only capture the pairwise relationship between check-ins and fail to explore higher order patterns. Although RUM strengthens the representation capacity compared with RNNs for sequential recommendation, it produces worse results than ST-RNN under some metrics. SASRec and AttRec outperform other baselines, which further validates the advantage of SAN in modeling user sequential behaviors.

C. Influence of Data Sparsity

In this section, we conduct extensive experiments to investigate the capacity of MAHAN in dealing with data sparsity. Due to the space limit, we only present the results in Acc@20 and NDCG@20. Experimental results in other metrics almost have similar trends and are omitted.

1) *Performance on Different Numbers of Historical Check-Ins*: To gain a deeper understanding of each method, we divide users into five groups according to the number of historical check-ins: $H^u < 20$, $H^u \in [20, 50)$, $H^u \in [50, 100)$, and $H^u \geq 100$. The performance of all the methods on different groups is shown in Fig. 4. We find that the performance of almost all methods shows a downward trend as the number of check-ins increases, but MAHAN achieves the best performance for all of the five groups. This indicates that our model has excellent applicability for users with different check-in frequencies. Among these deep learning-based baselines, ST-RNN performs relatively well for users with less check-ins (e.g., $H^u < 20$). It indicates that RNN-based methods favor short check-in sequence and cannot store long-term information well. AttRec always achieves better performance than other baselines with sufficient check-ins (e.g., $H^u \geq 50$).

2) *Performance on Different Ratio of Training Samples*: We further evaluate MAHAN and deep learning-based baseline methods under different proportions of training samples. The results are shown in Fig. 5. We find that the performance of all models decreases as the reduction of training data. However, our MAHAN model always achieves the best performance under different ratio of training data. Even with 40% training samples, the improvements of MAHAN over the second best result in Acc@20 are 10.9% and 10.4% on Foursquare and Gowalla data sets, respectively. This observation proves that MAHAN can cope with data sparsity problem very well. Among the baselines, the performance of ST-RNN drops significantly when reducing the amount of training data. The reason is that ST-RNN needs sufficient data to learn the time- and distance-specific model parameters associated with users’

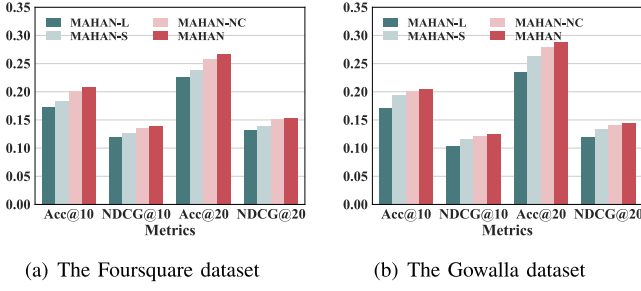


Fig. 6. Performance of MAHAN's variants. (a) Foursquare data set. (b) Gowalla data set.

transition patterns. AttRec still performs best among baselines, and this is mainly because it can simultaneously model long- and short-term preferences.

D. Effectiveness of Different Components in MAHAN

As shown above, MAHAN has obtained a significant performance improvement compared to the baselines, even under the sparse data sets. In this section, we compare MAHAN with its three variants to verify the effectiveness of different components. MAHAN-M and MAHAN-S denote that only the long-term module and the short-term module are preserved, respectively, whereas MAHAN-NC is a variant that removes the coattention network.

From the results given in Fig. 6, we can observe that the complete model MAHAN outperforms all the variants on both data sets, which demonstrates that each component of MAHAN makes positive contributions to the overall performance gain. MAHAN-S always performs better than MAHAN-M, which indicates that capturing sequential patterns is more conducive than modeling long-term preferences to the next POI prediction. Moreover, the performance of MAHAN-S is very close to SASRec and AttRec that also employ the self-attention-based sequential structure. This is mainly because our improved SAN can explore complicated spatiotemporal patterns within the check-in sequence. The integrated MAHAN-NC outperforms MAHAN-M and MAHAN-S. Besides, MAHAN-NC still outperforms all the baselines on both data sets. These results verify the necessity of combining long- and short-term preferences for prediction. MAHAN performs better than MAHAN-NC, and this suggests the effectiveness of the coattention network. Moreover, ST-SAN and KV-MN can complement and enhance each other well.

E. Detailed Analysis of MAHAN

In this section, we construct a detailed analysis of the ST-SAN and the memory network to better understand why and how they work.

1) *Ablation Study on Short-Term Module*: We implement two variants of ST-SAN to examine the effectiveness of ST-SAN in modeling temporal and spatial transition patterns. S-SAN only considers the spatial patterns, whereas T-SAN only learns the temporal patterns. The results are shown in Table IV. It is clear that the performance of S-SAN and T-SAN is better than SAN. Moreover, integrating two types

TABLE IV
PERFORMANCE OF DIFFERENT ST-SAN VARIANTS

Datasets	Methods	Acc@20	NDCG@20
Foursquare	SAN	0.2326	0.1349
	S-SAN	0.2365	0.1353
	T-SAN	0.2376	0.1384
	ST-SAN	0.2393	0.1397
Gowalla	SAN	0.2525	0.1279
	S-SAN	0.2599	0.1320
	T-SAN	0.2579	0.1295
	ST-SAN	0.2637	0.1336

TABLE V
PERFORMANCE OF THE VARIANTS WITH SHARED OR PRIVATE VALUE MATRICES FOR OUR MODEL

	Foursquare		Gowalla	
	Acc@20	NDCG@20	Acc@20	NDCG@20
Shared	0.2577	0.1498	0.2675	0.1369
Private	0.2662	0.1533	0.2889	0.1437

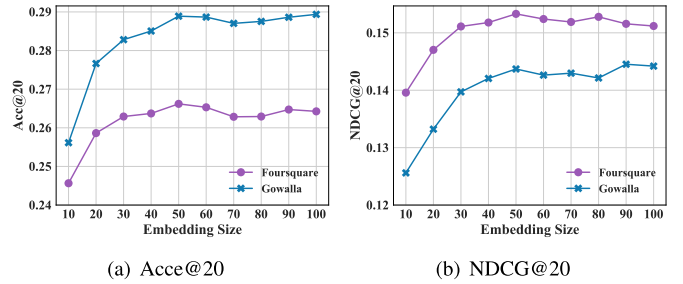


Fig. 7. Influence of embedding size d . (a) Acc@20. (b) NDCG@20.

of transition patterns achieves better results than using either pattern alone. This verifies exploring temporal and spatial patterns of the check-in sequence has positive impacts on precisely predicting a user's next destination.

2) *Personalized or Shared Value Matrix*: To verify the validity of maintaining dynamic and private value matrix in memory network for each user, we implement a variant of MAHAN with a static value matrix shared by all users. Table V presents the results. It is obvious that using the dynamic and private value matrix gets better performance. This confirms that the interests of each user own a unique evolving process. It is worth noting that the variant with shared value matrix still exceeds all the baselines. The possible reason is that the coattention network can generate dynamic and personal attention weights for the value matrix based on short-term preferences and target POIs. Thus, we can still model individual user preferences even when the value matrix is shared.

F. Influence of Hyperparameters

1) *Influence of Embedding Size d* : The performance of MAHAN with different embedding sizes d is shown in Fig. 7. With the increase of d , the performance of MAHAN gradually improves first and then becomes relatively stable. Moreover, a larger d does not necessarily lead to better performance. It is because the larger d has stronger expressive ability, but it increases the complexity of the model and leads to overfitting

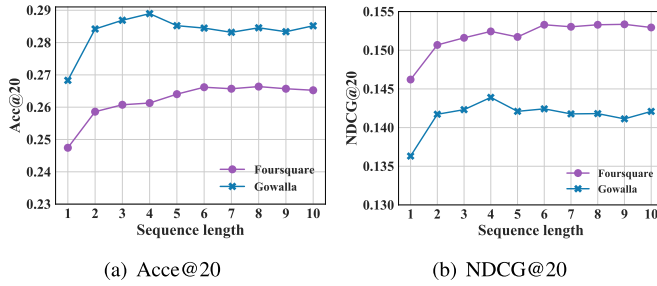


Fig. 8. Influence of sequence length k . (a) Acc@20. (b) NDCG@20.

TABLE VI
MODEL PERFORMANCE WITH RESPECT TO DIFFERENT
NUMBERS OF MEMORY SLOTS

m	Foursquare		Gowalla	
	Acc@20	NDCG@20	Acc@20	NDCG@20
1	0.2592	0.1474	0.2709	0.1377
5	0.2662	0.1533	0.2760	0.1411
10	0.2642	0.1521	0.2889	0.1437
15	0.2657	0.1518	0.2803	0.1427
20	0.2650	0.1525	0.2770	0.1425

problems. Considering the tradeoff between the computation cost and the recommendation performance, we choose $d = 50$ for both data sets.

2) *Influence of Sequence Length k* : Fig. 8 shows the performance of MAHAN with different sequence lengths. We can observe that the model performance is gradually improved and tends to be stable on the Foursquare. However, the performance on Gowalla shows some improvement at the beginning as k increases, while larger k leads to degradation of the performance to some extent. The reason is that users' transition patterns involve a short sequence and a longer sequence will introduce noisy behavioral patterns to the model

3) *Influence of Memory Slot Number m* : The number of memory slots m determines the expressiveness and capacity of the memory network. Thus, we conduct experiments on MAHAN with different m values. Table VI shows that increasing the number of memory slots can improve the model performance to a certain degree. These results verify the effectiveness of modeling fine-grained user preferences on multiple latent aspects. Our MAHAN model achieves the best results at $m = 5$ and $m = 10$ for Foursquare and Gowalla, respectively.

VI. SUMMARY AND FUTURE WORK

In this article, we introduce MAHAN, a novel model for next POI recommendation. In particular, MAHAN is a unified hybrid model that incorporates ST-SAN and KV-MN. Besides, a novel coattention network is proposed to fully learn the inherent association and dynamic interaction between long- and short-term preferences. We conduct extensive experiments on two real-world LBSN data sets. The experimental results demonstrate that MAHAN significantly outperforms the state-of-the-art methods in terms of different evaluation metrics. Moreover, MAHAN has better ability to mitigate data sparsity problems than baselines. In the future, we will study how to incorporate more auxiliary information (e.g., categorical and

social information) of users and POIs to improve the flexibility and effectiveness of the proposed model.

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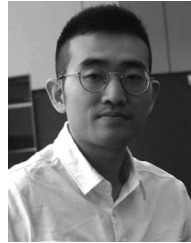
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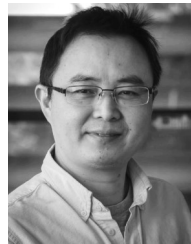
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